

HW13 GraphQL

Questions

1. What is GraphQL?
2. What are the features of GraphQL?
3. What is the difference between GraphQL and REST?
4. What is overfetching?
5. What is a GraphQL schema?
6. List the data types in GraphQL.
7. What are scalar types in GraphQL?
8. List the operations in GraphQL and explain each of them.
9. What are GraphQL variables and how do you use them?
10. What is the difference between using variables and hardcoding values directly inside the query?
11. Describe how you would fetch data with a GraphQL query.
12. Why are GraphQL requests usually sent with `POST` ?
13. Why does the `body` in `fetch` need `JSON.stringify(...)` ?
14. Why do we need to set `Content-Type: application/json` ?
15. What is the relationship between a GraphQL query and the JSON request body?
16. What is the difference between a network error and a GraphQL error?
17. How does GraphQL handle caching?
18. What is a persisted query?
19. What is the N+1 query issue in GraphQL?

Coding

1. Based on the JSON below

- a. Create GraphQL schema files.
- b. Write one configurable query to fetch the order by orderId, and return all nested data inside the order, including `customer`, `lineItems`, and each line item's `product`.
- c. Write down your answers - Explain `lineItems: [LineItem]!` `lineItems: [LineItem!]!` `lineItems: [LineItem!]!`

```
{
  "order": {
    "id": "ORD-001",
    "status": "SHIPPED",
    "createdAt": "2026-05-19T08:30:00Z",
    "customer": {
      "id": "C-100",
      "name": "Alice Smith",
      "email": "alice@example.com"
    },
    "lineItems": [
      {
        "id": "LI-001",
        "quantity": 2,
        "unitPrice": 79.99,
        "product": {
          "id": "P-500",
          "name": "Wireless Earbuds",
          "sku": "WEP-BLK",
          "category": "ELECTRONICS",
          "description": "Noise-cancelling earbuds with 24
h battery life"
        }
      },
      {
        "id": "LI-002",
        "quantity": 1,
        "unitPrice": 129.0,
```

```
    "product": {
      "id": "P-782",
      "name": "Running Shoes",
      "sku": "RSA-RED-42",
      "category": "SPORTS",
      "description": null
    }
  ],
  "total": 288.98
}
```

2. Create a React page that fetches Rick and Morty characters from the GraphQL API and renders them on the UI.
 - a. API - <https://rickandmortyapi.com/graphql>
 - b. Documentation - <https://rickandmortyapi.com/documentation>
 - c. The page must show for each character:
 - name
 - status
 - species
 - image
 - origin name
 - d. Minimum requirements:
 - Use `fetch`
 - Use GraphQL variables
 - Show loading state
 - Show error state
 - e. Bonus requirements:

- Pagination
- Search by name